

LAB EXPERIMENT RECORD

03-10-2026---Set 7: Customer Demographics Analysis

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Date	03 OCTOBER 2026
Day	Saturday

Aim

To analyze customer demographic data using a bar chart, pie chart and table, and to build an interactive Tableau dashboard for exploring customer demographics.

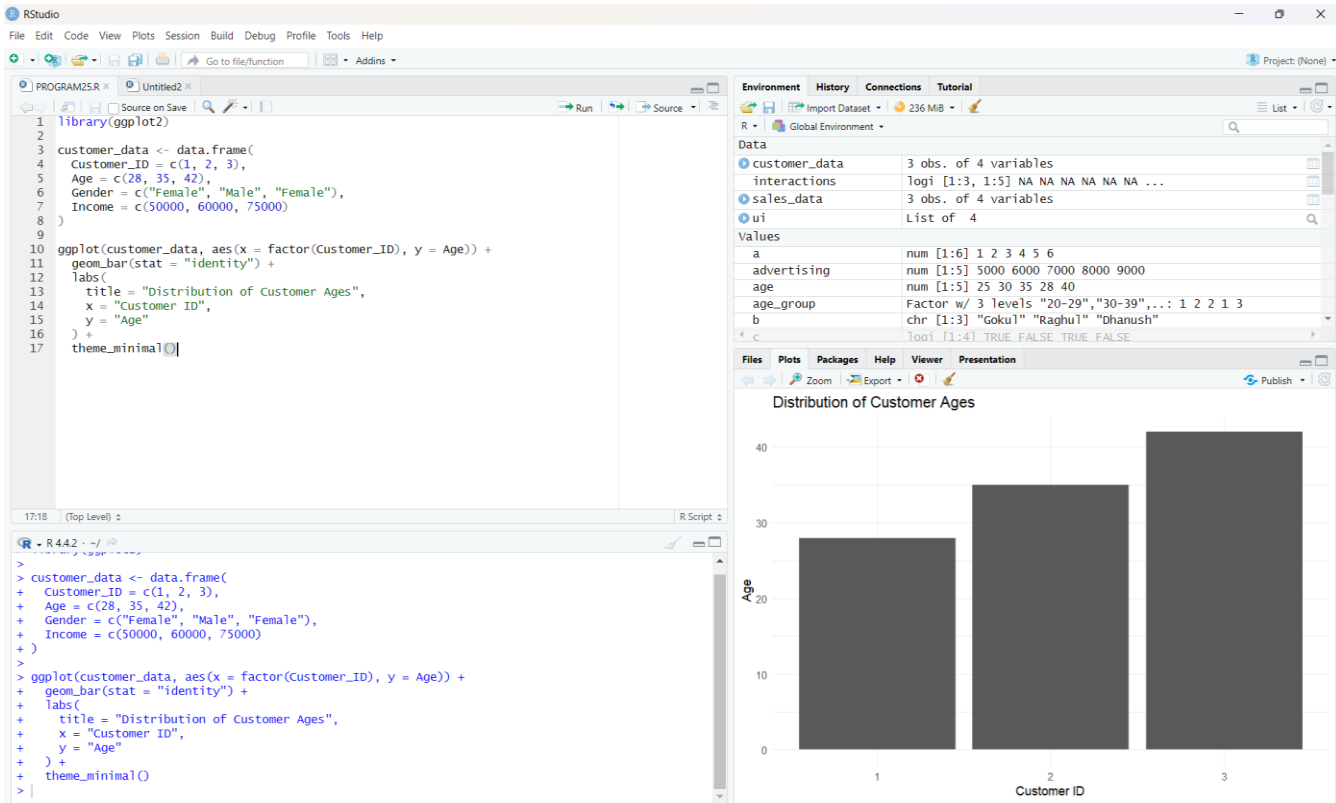
Dataset: Customer Demographics

Customer ID	Age	Gender	Income (in \$)
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

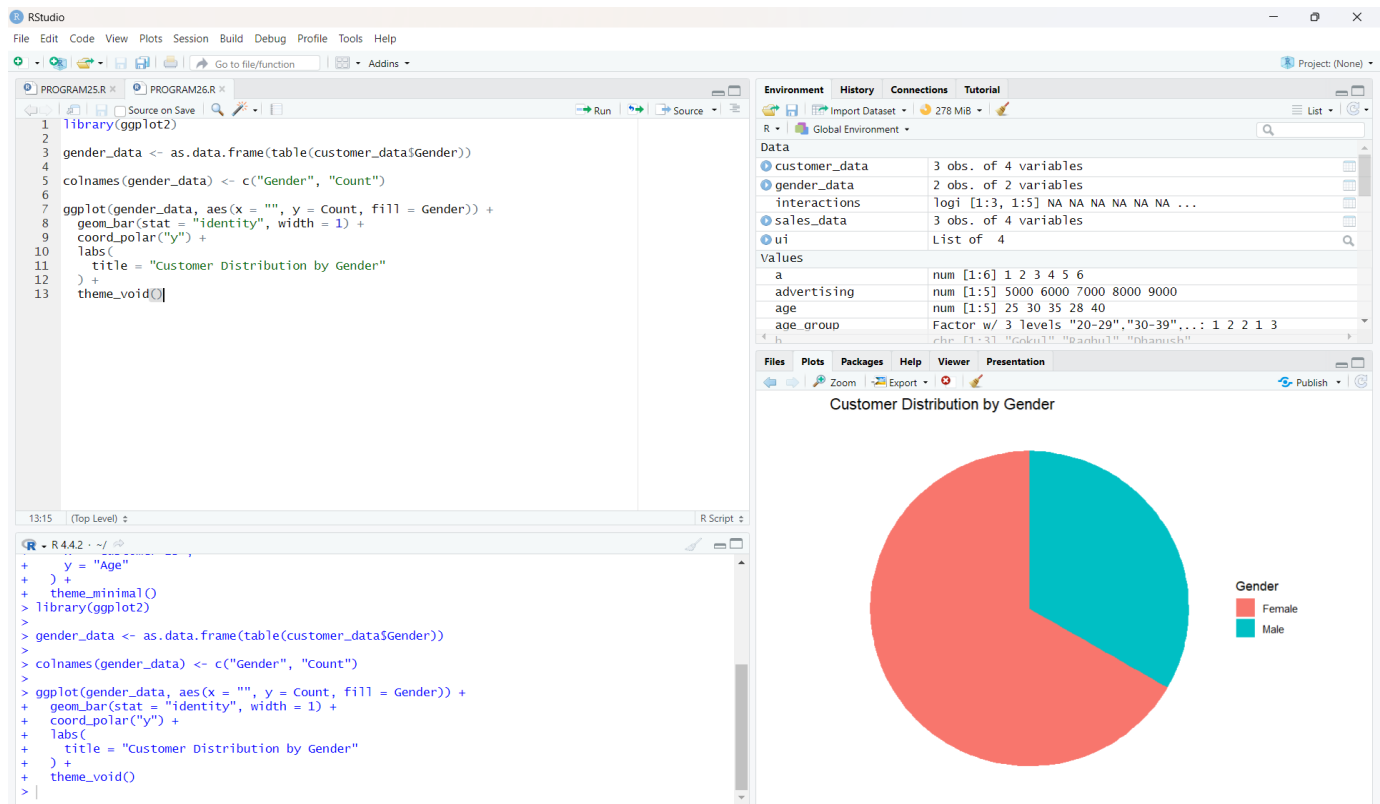
Question 1: Bar Chart of Customer Ages

Question: Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.

Aim: To visualize the distribution of customer ages using a bar chart with labelled axes and a title.



Result: The three customers are aged 28, 35 and 42, so the ages are spread evenly across a 14-year range with one customer at each age.

Question 2: Pie Chart of Customers by Gender**Question:** Generate a pie chart to represent the distribution of customers by gender.**Aim:** To represent the share of customers by gender using a labelled pie chart.**Result:** Female customers form 66.7% (2 customers) and male customers form 33.3% (1 customer).

Question 3: Table of Demographic Information

Question: Build a table to show the demographic information for each customer. Label the table elements.

Aim: To present the age, gender and income of each customer in a clearly labelled table.

The screenshot shows the RStudio interface. The script editor on the left contains the following R code:

```
1 customer_data <- data.frame(
2   Customer_ID = c(1, 2, 3),
3   Age = c(28, 35, 42),
4   Gender = c("Female", "Male", "Female"),
5   Income = c(50000, 60000, 75000)
6 )
7
8 print(customer_data)
```

The console at the bottom shows the output of the code:

```
> print(customer_data)
  Customer_ID Age Gender Income
1          1  28 Female 50000
2          2  35  Male 60000
3          3  42 Female 75000
```

The Environment pane on the right shows the objects created in the Global Environment:

- customer_data**: 3 obs. of 4 variables
- gender_data**: 2 obs. of 2 variables
- interactions**: logi [1:3, 1:5] NA NA NA NA NA ...
- sales_data**: 3 obs. of 4 variables
- ui**: List of 4

The Values pane shows the data for the selected object:

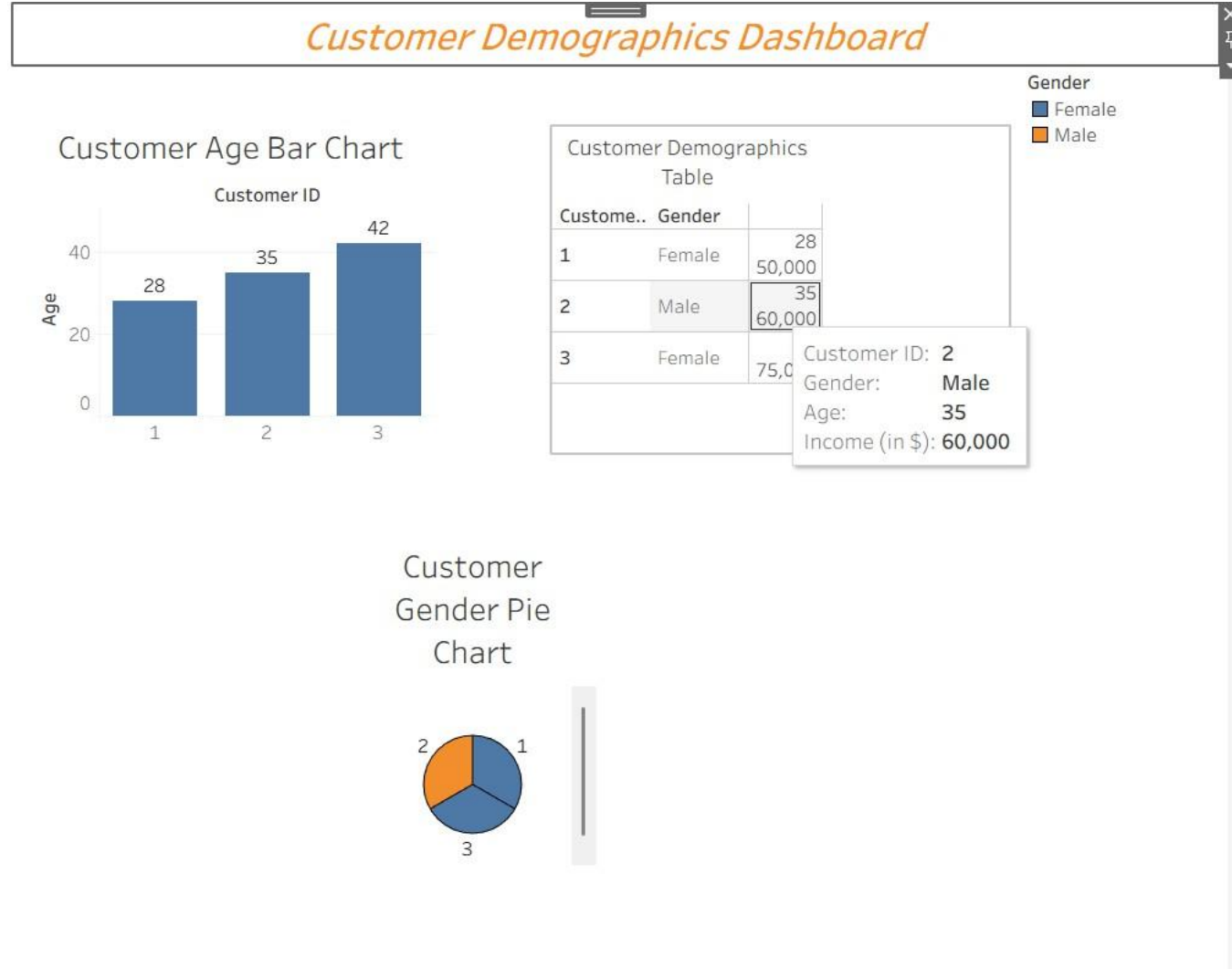
Variable	Values
a	num [1:6] 1 2 3 4 5 6
advertising	num [1:5] 5000 6000 7000 8000 9000
age	num [1:5] 25 30 35 28 40
age_group	Factor w/ 3 levels "20-29","30-39"...: 1 2 2 1 3

Result: The table lists all three customers with their details; Customer 3 has the highest income (75000) and Customer 1 the lowest (50000).

Question 4: Interactive Tableau Dashboard

Question: Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive exploration of customer demographics.

Aim: To build an interactive Tableau dashboard that combines the bar chart, pie chart and table for exploring customer demographics.



Result: The dashboard places all three views together, and a gender filter updates every view so users can explore the demographics interactively.

Overall Result

Observation: The average customer age is 35 years and the average income is about 61,667, with income rising as age increases.